Fall Guardian — Cloud Gateway 0.2.0

OAS 3.1

/openapi.json

Authorize

health

GET /health Health

Parameters Cancel

No parameters

Execute

Responses

Code	Description	Links
200	Successful Response	*No links*

Media type

application/json

Controls `Accept` header.

Example Value | Schema

```
{
  "status": "string",
  "version": "string",
  "model_version": "string",
  "environment": "string"
}
```

auth

POST `/v1/auth/register` Register

Parameters | Try it out

No parameters

Request body required | application/json

Example Value | Schema

```
{
  "email": "string",
  "password": "stringst",
  "full_name": "string"
}
```

Responses

Code	Description	Links
201	Successful Response Media type: application/json Controls `Accept` header. **Example Value** | Schema	*No links*

```
{
  "access_token": "string",
  "token_type": "bearer",
  "expires_in": 0
}
```

Code	Description	Links
422	Validation Error Media type: application/json **Example Value** | Schema	*No links*

```
{
  "detail": [
    {
      "loc": [
        "string",
        0
      ],
      "msg": "string",
      "type": "string",
      "input": "string",
      "ctx": {}
    }
```

Code	Description	Links

```
  ]
}
```

POST `/v1/auth/login` Login

Parameters | Try it out

No parameters

Request body required | application/json

Example Value | Schema

```
{
  "email": "string",
  "password": "string"
}
```

Responses

Code	Description	Links
200	Successful Response	*No links*

Media type: application/json

Controls `Accept` header.

Example Value | Schema

```
{
  "access_token": "string",
  "token_type": "bearer",
  "expires_in": 0
}
```

422	Validation Error	*No links*

Media type: application/json

Example Value | Schema

```
{
  "detail": [
    {
      "loc": [
        "string",
```

Code	Description	Links


```
        0
      ],
      "msg": "string",
      "type": "string",
      "input": "string",
      "ctx": {}
    }
  ]
}
```


inference

POST `/v1/inference` Inference

Parameters | Try it out

No parameters

Request body required | application/json

Example Value | Schema


```
{
  "device_id": "string",
  "ts_start_unix_ms": 0,
  "sample_rate_hz": 50,
  "samples": [
    {
      "ax": 0,
      "ay": 0,
      "az": 0,
      "wx": 0,
      "wy": 0,
      "wz": 0
    }
  ],
  "edge_prediction": {
    "p_pre_impact": 1,
    "model_version": "string"
  },
  "payload_type": "emergency"
}
```


Responses

Code	Description	Links
200	Successful Response Media type	*No links*

Code	Description	Links

application/json

Controls `Accept` header.

Example Value | Schema


```
{
  "is_fall": true,
  "confidence": 1,
  "severity": "none",
  "action": "string",
  "lead_time_ms": 0,
  "model_version": "string"
}
```


422 Validation Error — *No links*

Media type

application/json

Example Value | Schema


```
{
  "detail": [
    {
      "loc": [
        "string",
        0
      ],
      "msg": "string",
      "type": "string",
      "input": "string",
      "ctx": {}
    }
  ]
}
```


retraining

POST `/v1/retraining` Retraining

Parameters Try it out

No parameters

Request body required application/json

Example Value | Schema


```
{
  "device_id": "string",
  "ts_start_unix_ms": 0,
  "sample_rate_hz": 50,
  "samples": [
    {
      "ax": 0,
```

```
      "ay": 0,
      "az": 0,
      "wx": 0,
      "wy": 0,
      "wz": 0
    }
  ],
  "edge_prediction": {
    "p_pre_impact": 1,
    "model_version": "string"
  },
  "payload_type": "retraining_data"
}
```


Responses

Code	Description	Links
200	Successful Response Media type application/json Controls `Accept` header. **Example Value** | Schema	*No links*
422	Validation Error Media type application/json **Example Value** | Schema	*No links*


```
{
  "stored": true,
  "label": "string",
  "sample_id": "string",
  "message": "string"
}
```



```
{
  "detail": [
    {
      "loc": [
        "string",
        0
      ],
      "msg": "string",
      "type": "string",
      "input": "string",
      "ctx": {}
    }
  ]
}
```


events

GET /v1/events List Events

Parameters Try it out

Name	Description
device_id string | (string | null) *(query)*	filter to one device's events device_id
limit integer *(query)*	*Default value* : 50 50 *maximum: 200* *minimum: 1*
offset integer *(query)*	*Default value* : 0 0 *minimum: 0*

Responses

Code	Description	Links
200	Successful Response Media type application/json Controls `Accept` header. **Example Value** | Schema	*No links*


```
{
  "items": [
    {
      "id": "3fa85f64-5717-4562-b3fc-2c963f66afa6",
      "device_ref": "string",
      "ts_start_unix_ms": 0,
      "is_fall": true,
      "confidence": 0,
      "severity": "none",
      "lead_time_ms": 0,
      "model_version": "string",
      "acknowledged_at": "2026-06-05T07:17:12.260Z",
      "acked_by": "3fa85f64-5717-4562-b3fc-2c963f66afa6",
      "created_at": "2026-06-05T07:17:12.260Z"
    }
  ],
  "total": 0,
  "limit": 0,
```

Code	Description	Links

```
  "offset": 0
}
```

422 Validation Error — *No links*

Media type: application/json

Example Value | Schema

```
{
  "detail": [
    {
      "loc": [
        "string",
        0
      ],
      "msg": "string",
      "type": "string",
      "input": "string",
      "ctx": {}
    }
  ]
}
```

POST `/v1/events/{event_id}/acknowledge` Acknowledge Event

Parameters | Try it out

Name	Description
event_id * required `string($uuid)` *(path)*	event_id

Responses

Code	Description	Links
200	Successful Response	*No links*

Media type: application/json

Controls `Accept` header.

Example Value | Schema

```
{
  "id": "3fa85f64-5717-4562-b3fc-2c963f66afa6",
```

Code	Description	Links


```
  "device_ref": "string",
  "ts_start_unix_ms": 0,
  "is_fall": true,
  "confidence": 0,
  "severity": "none",
  "lead_time_ms": 0,
  "model_version": "string",
  "acknowledged_at": "2026-06-05T07:17:12.262Z",
  "acked_by": "3fa85f64-5717-4562-b3fc-2c963f66afa6",
  "created_at": "2026-06-05T07:17:12.262Z"
}
```


422 Validation Error *No links*

Media type

application/json

Example Value | Schema


```
{
  "detail": [
    {
      "loc": [
        "string",
        0
      ],
      "msg": "string",
      "type": "string",
      "input": "string",
      "ctx": {}
    }
  ]
}
```


GET `/v1/events/stream` Stream Events

Server-Sent Events feed of the caller's confirmed falls (Redis pub/sub).

Subscribes to the caller's own channel and relays each published alert as an SSE `fall` event; a comment frame every `_SSE_KEEPALIVE_S` keeps the link alive and surfaces client disconnects. 503 without Redis (see require_broker).

Parameters Try it out

No parameters

Responses

Code	Description	Links
200	Successful Response	*No links*

Media type

application/json

Controls `Accept` header.

Example Value | Schema

```
"string"
```

devices

POST `/v1/devices/pairing-codes` Create Pairing Code

Parameters Try it out

No parameters

Responses

Code	Description	Links
201	Successful Response	*No links*

Media type

application/json

Controls `Accept` header.

Example Value | Schema

```
{
  "code": "string",
  "expires_at": "2026-06-05T07:17:12.266Z"
}
```

POST `/v1/devices/pair` Pair

Parameters Try it out

No parameters

Request body required

application/json

Example Value | Schema

```
{
  "code": "stringst",
  "device_id": "string"
}
```

Responses

Code	Description	Links
200	Successful Response Media type: application/json Controls `Accept` header. **Example Value** | Schema `{ "device_token": "string", "token_type": "bearer", "device_id": "string", "user_id": "3fa85f64-5717-4562-b3fc-2c963f66afa6" }`	*No links*
422	Validation Error Media type: application/json **Example Value** | Schema `{ "detail": [ { "loc": [ "string", 0 ], "msg": "string", "type": "string", "input": "string", "ctx": {} } ] }`	*No links*

POST /v1/devices/heartbeat Heartbeat

Parameters Try it out

No parameters

Request body required application/json

Example Value | Schema

```
{
  "device_id": "string",
  "battery_pct": 100,
  "signal_dbm": 0,
  "edge_model_version": "string"
}
```

Responses

Code	Description	Links
200	Successful Response Media type: application/json Controls Accept header. Example Value | Schema	No links
422	Validation Error Media type: application/json Example Value | Schema	No links

```
{
  "id": "3fa85f64-5717-4562-b3fc-2c963f66afa6",
  "device_id": "string",
  "status": "string",
  "battery_pct": 0,
  "signal_dbm": 0,
  "last_seen_at": "2026-06-05T07:17:12.268Z",
  "paired_at": "2026-06-05T07:17:12.268Z",
  "edge_model_version": "string",
  "created_at": "2026-06-05T07:17:12.268Z"
}
```

```
{
  "detail": [
    {
      "loc": [
        "string",
```

Code	Description	Links


```
        0
      ],
      "msg": "string",
      "type": "string",
      "input": "string",
      "ctx": {}
    }
  ]
}
```


GET `/v1/devices` List Devices

Parameters Try it out

No parameters

Responses

Code	Description	Links
200	Successful Response Media type: application/json Controls `Accept` header. **Example Value** | Schema	*No links*


```
[
  {
    "id": "3fa85f64-5717-4562-b3fc-2c963f66afa6",
    "device_id": "string",
    "status": "string",
    "battery_pct": 0,
    "signal_dbm": 0,
    "last_seen_at": "2026-06-05T07:17:12.269Z",
    "paired_at": "2026-06-05T07:17:12.269Z",
    "edge_model_version": "string",
    "created_at": "2026-06-05T07:17:12.269Z"
  }
]
```


Schemas

DeviceOut Collapse all `object`

A device's live status. `status` is derived from `last_seen_at` at read time.

id* string uuid

device_id* string

status* string

battery_pct* ^ Collapse all (integer | null)

Any of ^ Collapse all (integer | null)

#0 integer

#1 null

signal_dbm* ^ Collapse all (integer | null)

Any of ^ Collapse all (integer | null)

#0 integer

#1 null

last_seen_at* ^ Collapse all (string | null)

Any of ^ Collapse all (string | null)

#0 string date-time

#1 null

paired_at* ^ Collapse all (string | null)

Any of ^ Collapse all (string | null)

#0 string date-time

#1 null

edge_model_version* ^ Collapse all (string | null)

Any of ^ Collapse all (string | null)

#0 string

#1 null

created_at* string date-time

EdgePrediction ^ Collapse all object

The edge model's trigger, included only when the edge fired.

p_pre_impact* number [0, 1]

model_version* string

EventOut ^ Collapse all object

A persisted fall verdict, as returned on the caregiver timeline.

id* string uuid

device_ref* string

ts_start_unix_ms* integer

is_fall* boolean

confidence* number

severity* ^ Collapse all string

Enum ^ Collapse all array

#0="none"

#1="low"

#2="medium"

#3="high"

lead_time_ms* ^ Collapse all (number | null)

Any of ^ Collapse all (number | null)

#0 number

#1 null

model_version* string

acknowledged_at* ^ Collapse all (string | null)

Any of ^ Collapse all (string | null)

#0 string date-time

#1 null

acked_by* ^ Collapse all (string | null)

Any of ^ Collapse all (string | null)

#0 string uuid

#1 null

created_at* string date-time

EventPage ^ Collapse all object

A page of the fall-event timeline (newest first).

items* ^ Collapse all array<object>

Items ^ Collapse all object

A persisted fall verdict, as returned on the caregiver timeline.

id* string uuid

device_ref* string

ts_start_unix_ms* integer

is_fall* boolean

confidence* number

severity* ^ Collapse all string

Enum ^ Collapse all array

#0="none"

#1="low"

#2="medium"

#3="high"

lead_time_ms* ^ Collapse all (number | null)

Any of ^ Collapse all (number | null)

#0 number

#1 null

model_version* string

acknowledged_at* ^ Collapse all (string | null)

Any of ^ Collapse all (string | null)

#0 string date-time

#1 null

acked_by* Collapse all (string | null)

Any of Collapse all (string | null)

#0 string uuid

#1 null

created_at* string date-time

total* integer

limit* integer

offset* integer

HTTPValidationError Collapse all object

detail Collapse all array<object>

Items Collapse all object

loc* Collapse all array<(string | integer)>

Items Collapse all (string | integer)

Any of Collapse all (string | integer)

#0 string

#1 integer

msg* string

type* string

input any

ctx object

HealthResponse Collapse all object

status* string

version* string

model_version* string

environment* string

HeartbeatRequest Collapse all object

A periodic device status ping (ARCHITECTURE §2.1 — watch sends ~every 5 min).

device_id* string ≥ 1 characters

battery_pct Collapse all (integer | null)

Any of Collapse all (integer | null)

#0 integer [0, 100]

#1 null

signal_dbm Collapse all (integer | null)

Any of Collapse all (integer | null)

#0 integer

#1 null

edge_model_version ^ Collapse all (string | null)

Any of ^ Collapse all (string | null)

#0 string

#1 null

IMUSample ^ Collapse all object

One IMU reading: 3-axis accel (m/s²) + 3-axis gyro (rad/s).

ax* number

ay* number

az* number

wx* number

wy* number

wz* number

InferenceRequest ^ Collapse all object

A 2.5 s window streamed to the cloud when the edge model fires (Payload A).
`payload_type` defaults to `emergency` so the locked §8 contract and existing clients (ESP32 firmware, virtual device) keep working without sending it.

device_id* string ≥ 1 characters

ts_start_unix_ms* integer ≥ 0

sample_rate_hz ^ Collapse all integer

Default=50

samples* ^ Collapse all array<object>

Items ^ Collapse all object

One IMU reading: 3-axis accel (m/s²) + 3-axis gyro (rad/s).

ax* number

ay* number

az* number

wx* number

wy* number

wz* number

edge_prediction ^ Collapse all (object | null)

Any of ^ Collapse all (object | null)

#0 EdgePrediction ^ Collapse all object

The edge model's trigger, included only when the edge fired.

p_pre_impact* number [0, 1]

model_version* string

#1 null

payload_type ^ Collapse all string

Why the watch uploaded this window — picks the cloud routing.

`emergency` → a live edge trigger; goes to the CloudDetector for secondary verification (and, later, caregiver alerting). `retraining_data` → a window the user CANCELED during the local grace period (false alarm); skips detection and is stored as labeled training data for future fine-tuning / threshold tuning.

Enum ^ Collapse all array

#0="emergency"

#1="retraining_data"

Default="emergency"

InferenceResponse ^ Collapse all object

The cloud detector's verdict on a window.

is_fall* boolean

confidence* number [0, 1]

severity* ^ Collapse all string

Enum ^ Collapse all array

#0="none"

#1="low"

#2="medium"

#3="high"

action* string

lead_time_ms ^ Collapse all (number | null)

Any of ^ Collapse all (number | null)

#0 number

#1 null

model_version* string

LoginRequest ^ Collapse all object

email* string

password* string

PairRequest ^ Collapse all object

code* string 8 characters

device_id* string ≥ 1 characters

PairResponse ^ Collapse all object

The device's long-lived token, returned once on successful pairing.

device_token* string

token_type ^ Collapse all string

Default="bearer"

device_id* string

user_id* string uuid

PairingCodeResponse ^ Collapse all object

A short-lived code a paired user shows to a device during provisioning.

code* string

expires_at* string date-time

PayloadType ^ Collapse all string

Why the watch uploaded this window — picks the cloud routing. `emergency` → a live edge trigger; goes to the CloudDetector for secondary verification (and, later, caregiver alerting). `retraining_data` → a window the user CANCELED during the local grace period (false alarm); skips detection and is stored as labeled training data for future fine-tuning / threshold tuning.

Enum ^ Collapse all array

#0="emergency"

#1="retraining_data"

RegisterRequest ^ Collapse all object

email* string [3, 320] characters

password* string [8, 128] characters

full_name ^ Collapse all (string | null)

Any of ^ Collapse all (string | null)

#0 string

#1 null

RetrainingAck ^ Collapse all object

Ack for a stored retraining sample. No detection verdict — by design.

stored* boolean

label* string

sample_id* string

message* string

RetrainingRequest ^ Collapse all object

A user-canceled false alarm uploaded for MLOps (Payload B). `payload_type` is pinned to `retraining_data`: posting an `emergency` window to the retraining endpoint is a 422, so a live trigger can't be silently diverted into the data-collection path. Clients may omit the field (the URL implies it).

device_id* string ≥ 1 characters

ts_start_unix_ms* integer ≥ 0

sample_rate_hz ^ Collapse all integer

Default=50

samples* ^ Collapse all array<object>

Items Collapse all object

One IMU reading: 3-axis accel (m/s²) + 3-axis gyro (rad/s).

ax* number

ay* number

az* number

wx* number

wy* number

wz* number

edge_prediction Collapse all (object | null)

Any of Collapse all (object | null)

#0 EdgePrediction Collapse all object

The edge model's trigger, included only when the edge fired.

p_pre_impact* number [0, 1]

model_version* string

#1 null

payload_type Collapse all string

Const="retraining_data"

Default="retraining_data"

Severity Collapse all string

Enum Collapse all array

#0="none"

#1="low"

#2="medium"

#3="high"

TokenResponse Collapse all object

access_token* string

token_type Collapse all string

Default="bearer"

expires_in* integer

ValidationError Collapse all object

loc* Collapse all array<(string | integer)>

Items Collapse all (string | integer)

Any of Collapse all (string | integer)

#0 string

#1 integer

msg* string

type* string

```
input  any
ctx  object
```